Question Driven Instruction for Mathematics

David Hodge

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Origins

Peer Instruction comes out of the work of Eric Mazur in the 1990s.

It's a flipped classroom approach.

It lives inside the more generic area of using audience response systems, where you'll find the term Question Driven Instruction.

Mazur introduced the method to Physics teaching, specifically to concept learning.

Concept questions

Definition

Questions that can be answered based on knowledge of the concepts and ideas, without extensive calculation.

> *e.g. What would happen in this physical system ... (image)... if we move this electromagnet in that direction?*

Fundamental idea

It's about active learning

The central philosophy is to focus on the time spent in the classroom, making that as engaging and useful for learning / information assimilation as possible.

> ⚠ **Key Idea**
>
> Focussing time in the classroom on trying to more deeply learn concepts and ideas, and not on just receiving information to learn later.

The process

- Pose question
- Students think about it (individually) ⟸ TIME
- Poll
- (Discussion by you)
- Discussion between students ⟸ TIME
- Poll again
- Explanation

Software requirements

Typically best to use software to streamline the polling process, I've used

- Pingo (https://pingo.coactum.de)
- Mentimeter
- ~~Itempool (online)~~

Many, many more options.

Tiny Pingo Demo

https://pingo.coactum.de/113580



Static Link to Session

Staff link: https://pingo.coactum.de/events/113580

(Try OPT Ch0, (5,1) example) or (Try L11MAF)

Example questions

Example

Construct a function such that … (word clouds)

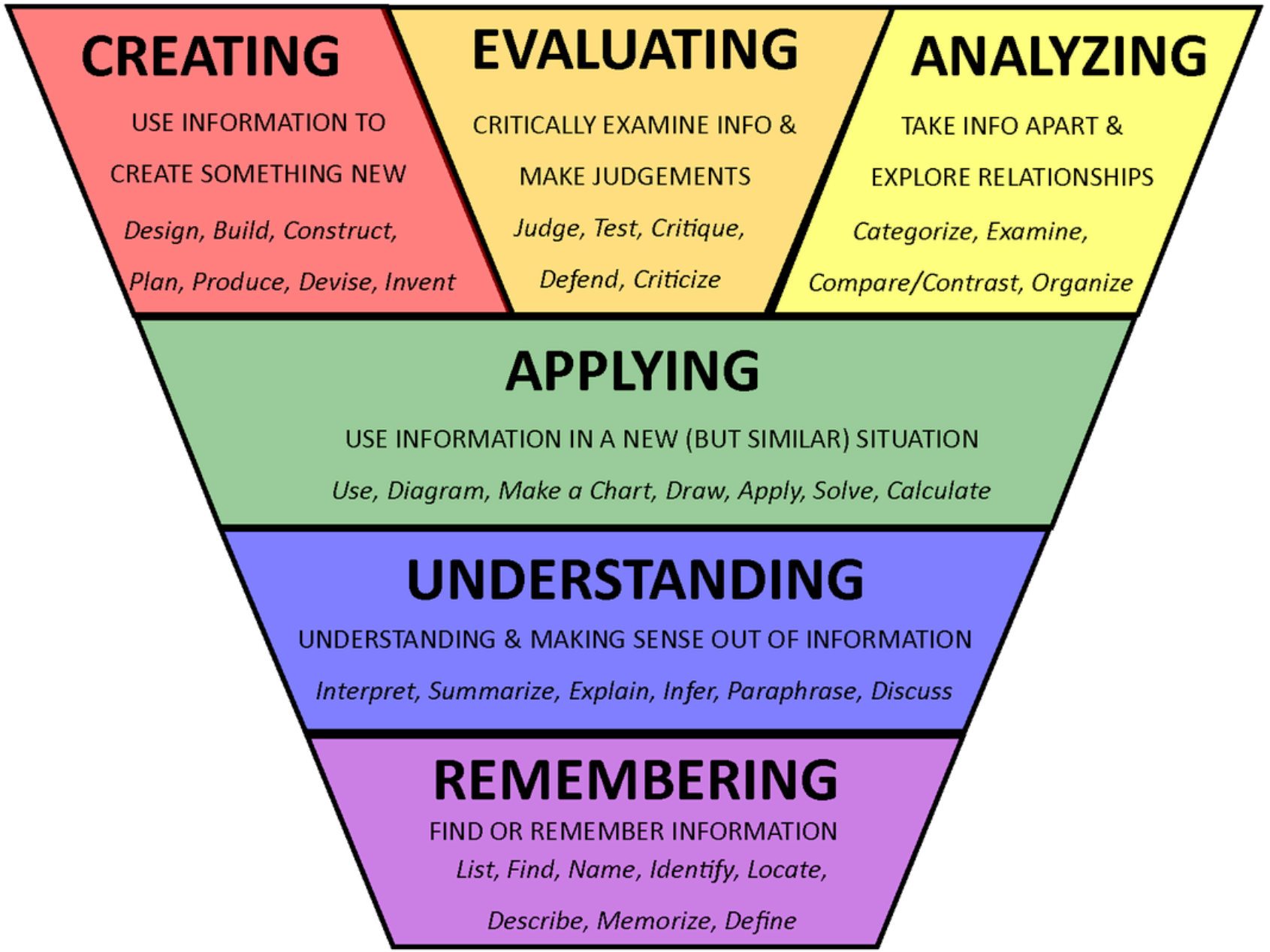
Example

What happens to the solution of … when we change … (big MCQ)

Example

What's the key difference between X and Y? (big MCQ)

I have mostly designed Multiple Choice Questions (MCQs) with multiple correct answers. Correctness often not important.



Bloom's Taxonomy

What make good questions?

Typical answers

- Good distractors
- Encouraging ‘linking’ knowledge
- Non-computational (numerical errors)
- Building upon knowledge, not recalling
- Target higher order thinking

I also sometimes like ambiguous questions (e.g special cases).

Promoting discussion (though trying not to confuse).

Start of term guidance

When using this kind of approach, what should we tell students to help them get the most from it?

- Spend time reading guidance (with them)
- Explain the focus on learning not testing
- Explain lecture format
- Explain evidence of effectiveness and where benefits are

Smaller alternatives

- Recap quizzes
- Warm-up questions

- Promoting group learning
- Progress/understanding check
- Taking students' questions

- End of class recap/check
- Exit questions/reflections/intentions

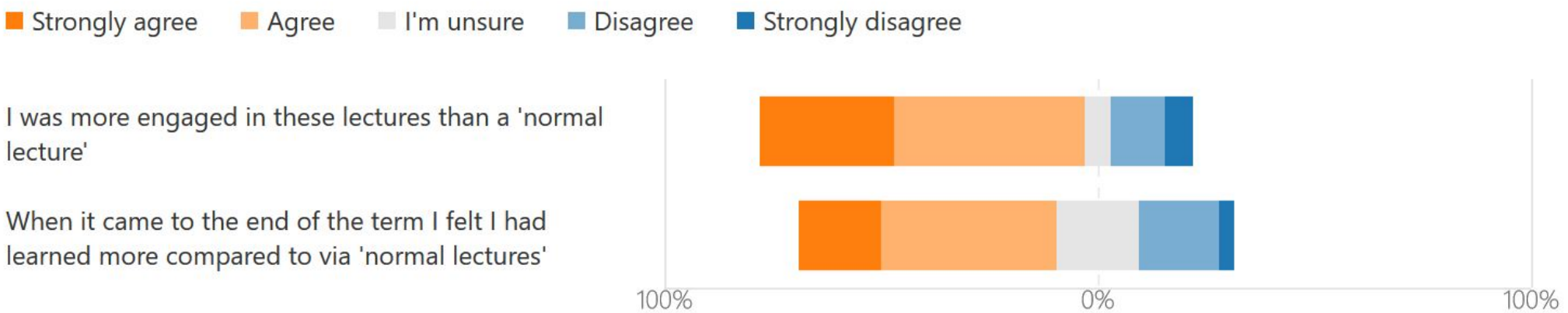
Thanks!

A few bonus slides

Detailed structure (Year 3 Math Finance)

- Pre-lecture reading and optional exercises;
- Lecture content: three or four challenging questions;
- Lecture format, per question:
  - Individual thought then audience response;
  - Lecturer guidance;
  - Small group discussions, then audience response;
  - Lecturer guided discussion;
- Class size was 160 students, in a level 3 module.
- Lecture time focused upon student understanding, thinking, and learning, not note-taking.

Engagement



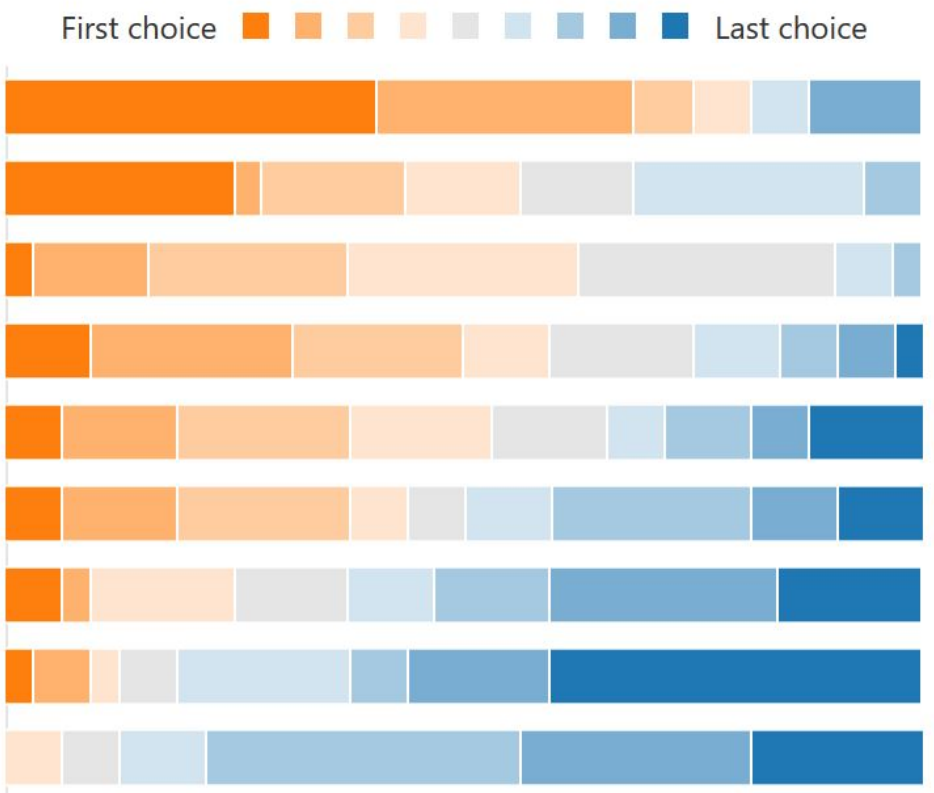
Engagement

Usefulness rankings

Please these elements (approximately) in order, in terms of your view of their helpfulness for your learning (not revision). MOST HELPFUL AT THE TOP. (Drag-and-drop is available)

More Details

Rank	Options
1	The lecture notes
2	The questions themselves
3	The explanations of the answers
4	Past papers
5	The homework exercises
6	The discussion time in lectures...
7	The Study Plan grid
8	Lecture video recordings
9	Office hours



Usefulness Rankings

Language advice

- We want students to ask “How do I work this out?” and “What am I missing?” and not “Why did the lecturer not teach me this?”. Using a less memory-based open question style can help.
- Talk about learning!

> “
>
> *Thinking about this in terms of xxx... is one popular way to help you learn it.*
>
> *Some of you may find the similarity with xxx... a useful method to help you learn this.*